



Python for Data Science - 2305CS303

Lab - 15

Roll No. : 122

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✓ Given an image file:

```
from PIL import Image
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

✓ 01) Read the image using the PIL library.

```
img1 = Image.open("download.jfif")
```

```
plt.imshow(img1)  
plt.show()
```



```
plt.imshow(img1)  
plt.axis("off")  
plt.show()
```



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- ✓ 02) Convert the image into a NumPy array.

```
img1_array = np.array(img1)
```

```
print(img1_array.shape)
```

```
(675, 1705, 3)
```

- ✓ 03) Create a copy of the image array.

```
img_copy = img1_array.copy()
```

- ✓ 04) Extract a sub-image starting from the 100th row and 100th column, selecting every 20th row and column.

```
subimg = img_copy[100::20, 100::20]
```

- ✓ 05) Display the processed image using Matplotlib.

```
plt.imshow(subimg)  
plt.axis("off")  
plt.show()
```



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